Experiments for Project1

Assumptions

L1 Associativity (S1): I limited the search for S1 to a maximum of **4 (16-way)**. While the simulator allows higher values, real-world L1 caches typically do not exceed 16-way due to high power and area costs. Furthermore, every increment in S1 increases hit time by 0.2 ns, leading to diminishing returns where latency outweighs miss-rate benefits.

L2 Associativity (S2): I excluded **S2=0** (direct-mapped) to ensure L2 robustness against conflict misses and to comply with the architectural requirement that **S2>=S1** The search was extended to **S2=8** to explore full capacity for mitigating conflict misses in memory-intensive traces like matmul_naive.

The design space was constrained by fundamental cache principles. We limited the block size to 128B because, while **larger block sizes** can improve performance by exploiting **spatial locality**, they also risk **cache pollution** by bringing in unnecessary data that displaces useful blocks.

Methodology and Performance Evaluation

The results were generated by executing ran_experiments.py, which automated a design space sweep across all six benchmarks to calculate the **L1 Average Access Time (AAT)** for each configuration. **L1 Average Access Time (AAT)** served as the primary metric, effectively balancing miss-rate reductions against the hit-time penalties of higher associativity.

My selection logic prioritized the lowest **AAT**. In cases of functional equivalence (within 0.001 ns), I selected the configuration with the **minimal metadata footprint**. This ensures the recommended settings maximize hardware efficiency and minimize the area and power costs of tag arrays and predictor tables.

In cases where designs remained tied across both performance and metadata, **MIP (Most Recently Used Insertion Policy)** would be selected. While both MIP and

LIP performed identically in these instances, MIP was chosen because it is the industry-standard default for robust set-associative caches.

I also decided to keep the **L2 cache** enabled for all configurations to ensure the system remained performant. Without it, the high latency of main memory would dominate the AAT, making L1-specific optimizations nearly invisible in the final results.

To identify the point of **diminishing returns** for associativity (S1 and S2) against **AAT**, I look for the "elbow" in the curve where adding more associativity either stops helping or actually starts hurting your performance.

Experiment Configurations

C1 = [14, 15] (16KB - 32KB)

C2 = [16, 17] (64KB - 128KB)

B = [5, 6, 7] (32B - 128B)

S1 = [0, 1, 2, 3, 4]

S2 = [1, 2, 3, 4, 8]

L2_POLICIES = ["mip", "lip"]

PREFETCHERS = ["none", "plus1", "markov", "hybrid"]

MARKOV_ROWS = [0, 16, 32, 64, 128, 256, 512]

Experiment Results

GCC Trace

Prioritizing **AAT** for the gcc benchmark, I found **4** configurations giving the lowest AAT of **2.536**

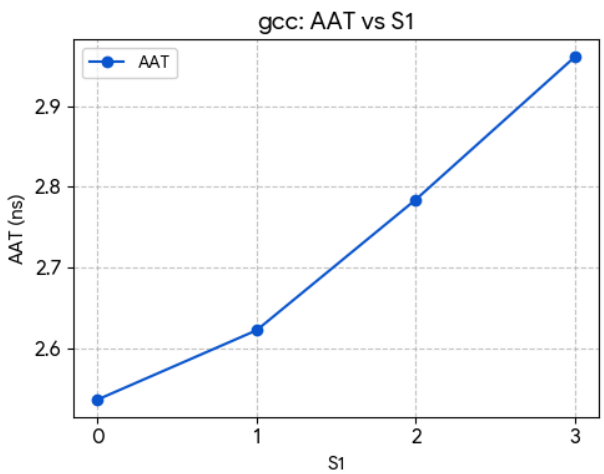
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	**Trace**	**C1**	**B**	**S1**	**C2**	**B.1**	**S2**	**L2 Policy**	**Prefetch**	**Rows**	**AAT**	**Data_B**	**Meta_B**
2	gcc	15	7	0	17	7	3	mip	hybrid	256	2.536	163840	66304
3	gcc	15	7	0	17	7	3	mip	hybrid	512	2.536	163840	66304
4	gcc	15	7	0	17	7	3	lip	hybrid	256	2.536	163840	66304
5	gcc	15	7	0	17	7	3	lip	hybrid	512	2.536	163840	66304

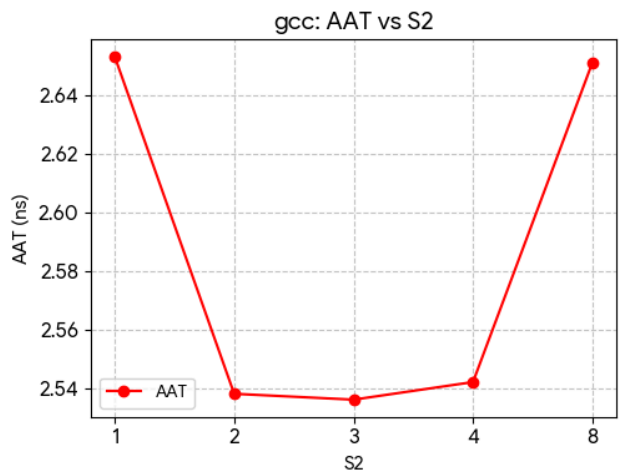
Selection Rationale

Since metadata storage was identical across these winners, the final selection was based on architectural simplicity. I selected the **hybrid prefetcher with 256 rows** over the 512-row version, as doubling capacity provided no additional performance benefit. This choice follows the principle of diminishing returns by opting for the lower hardware complexity of a smaller predictor table, as increasing the table to 512 rows yielded no additional performance gain.

For the **gcc** benchmark, the best configuration is **L1(15, 7, 0) L2(17, 7, 3), MIP policy** for L2 and **Hybrid prefetcher** with **256 rows**, achieving an AAT of **2.536**.

Diminishing Returns





The point of diminishing returns for S1 is immediately at **S1 = 0.** For the L2 cache, the curve follows a more traditional "U-shape". The point of diminishing returns for S2 is **S2 = 3 (8-way)**

Leela Trace

For the leela benchmark, I identified **28** configurations that achieved the minimum AAT of **2.078**

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	**Trace**	**C1**	**B**	**S1**	**C2**	**B.1**	**S2**	**L2 Policy**	**Prefetch**	**Rows**	**AAT**	**Data_B**	**Meta_B**
2	leela	15	7	0	16	7	1	mip	plus1	0	2.078	98304	39168
3	leela	15	7	0	16	7	1	mip	hybrid	16	2.078	98304	39168
4	leela	15	7	0	16	7	1	mip	hybrid	32	2.078	98304	39168
5	leela	15	7	0	16	7	1	mip	hybrid	64	2.078	98304	39168
6	leela	15	7	0	16	7	1	mip	hybrid	128	2.078	98304	39168
7	leela	15	7	0	16	7	1	mip	hybrid	256	2.078	98304	39168
8	leela	15	7	0	16	7	1	mip	hybrid	512	2.078	98304	39168
9	leela	15	7	0	16	7	1	lip	plus1	0	2.078	98304	39168
10	leela	15	7	0	16	7	1	lip	hybrid	16	2.078	98304	39168
11	leela	15	7	0	16	7	1	lip	hybrid	32	2.078	98304	39168
12	leela	15	7	0	16	7	1	lip	hybrid	64	2.078	98304	39168
13	leela	15	7	0	16	7	1	lip	hybrid	128	2.078	98304	39168
14	leela	15	7	0	16	7	1	lip	hybrid	256	2.078	98304	39168
15	leela	15	7	0	16	7	1	lip	hybrid	512	2.078	98304	39168
16	leela	15	7	0	17	7	1	mip	plus1	0	2.078	163840	64256
17	leela	15	7	0	17	7	1	mip	hybrid	16	2.078	163840	64256
18	leela	15	7	0	17	7	1	mip	hybrid	32	2.078	163840	64256
19	leela	15	7	0	17	7	1	mip	hybrid	64	2.078	163840	64256
20	leela	15	7	0	17	7	1	mip	hybrid	128	2.078	163840	64256
21	leela	15	7	0	17	7	1	mip	hybrid	256	2.078	163840	64256
22	leela	15	7	0	17	7	1	mip	hybrid	512	2.078	163840	64256
23	leela	15	7	0	17	7	1	lip	plus1	0	2.078	163840	64256
24	leela	15	7	0	17	7	1	lip	hybrid	16	2.078	163840	64256
25	leela	15	7	0	17	7	1	lip	hybrid	32	2.078	163840	64256
26	leela	15	7	0	17	7	1	lip	hybrid	64	2.078	163840	64256
27	leela	15	7	0	17	7	1	lip	hybrid	128	2.078	163840	64256
28	leela	15	7	0	17	7	1	lip	hybrid	256	2.078	163840	64256
29	leela	15	7	0	17	7	1	lip	hybrid	512	2.078	163840	64256

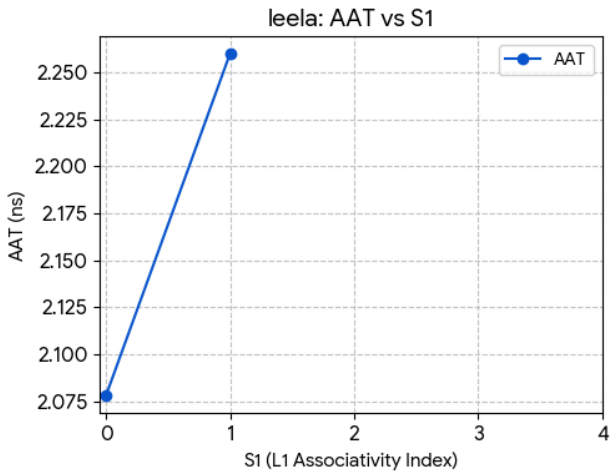
Applying the selection logic of lowest cache costs (smallest metadata and hardware simplicity), the optimal design is:

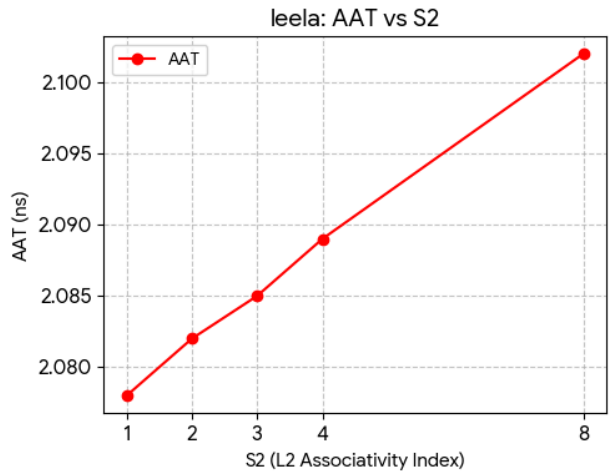
For the **leela** benchmark, the best configuration is **L1(15, 7, 0) L2(16, 7, 1)**, **MIP policy** and **Plus1 prefetcher**, achieving an AAT of **2.078**

Selection Rationale

Metadata and Simplicity: While several configurations shared the same metadata footprint (39,168 bits), the Plus1 prefetcher was selected over the Hybrid model as it achieves the peak performance with lower architectural complexity.

Diminishing Returns





The point of diminishing returns for S1 is reached at **S1 = 0**. The point of diminishing returns for S2 is reached at **S2 = 1 (2-way)**. While increasing associativity typically helps resolve conflict misses, the leela trace is already highly efficient at 2-way. Adding more ways (and thus adding 0.2 ns of latency per step) consistently makes the AAT worse.

Linpack Trace

For the linpack benchmark, I identified **14** configurations yielding the lowest AAT of **4.523**

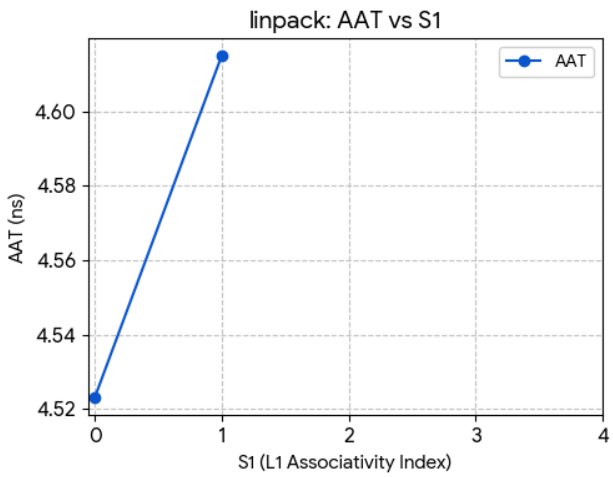
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	**Trace**	**C1**	**B**	**S1**	**C2**	**B.1**	**S2**	**L2 Policy**	**Prefetch**	**Rows**	**AAT**	**Data_B**	**Meta_B**
2	linpack	15	7	0	17	7	1	mip	plus1	0	4.523	163840	64256
3	linpack	15	7	0	17	7	1	mip	hybrid	16	4.523	163840	64256
4	linpack	15	7	0	17	7	1	mip	hybrid	32	4.523	163840	64256
5	linpack	15	7	0	17	7	1	mip	hybrid	64	4.523	163840	64256
6	linpack	15	7	0	17	7	1	mip	hybrid	128	4.523	163840	64256
7	linpack	15	7	0	17	7	1	mip	hybrid	256	4.523	163840	64256
8	linpack	15	7	0	17	7	1	mip	hybrid	512	4.523	163840	64256
9	linpack	15	7	0	17	7	1	lip	plus1	0	4.523	163840	64256
10	linpack	15	7	0	17	7	1	lip	hybrid	16	4.523	163840	64256
11	linpack	15	7	0	17	7	1	lip	hybrid	32	4.523	163840	64256
12	linpack	15	7	0	17	7	1	lip	hybrid	64	4.523	163840	64256
13	linpack	15	7	0	17	7	1	lip	hybrid	128	4.523	163840	64256
14	linpack	15	7	0	17	7	1	lip	hybrid	256	4.523	163840	64256
15	linpack	15	7	0	17	7	1	lip	hybrid	512	4.523	163840	64256

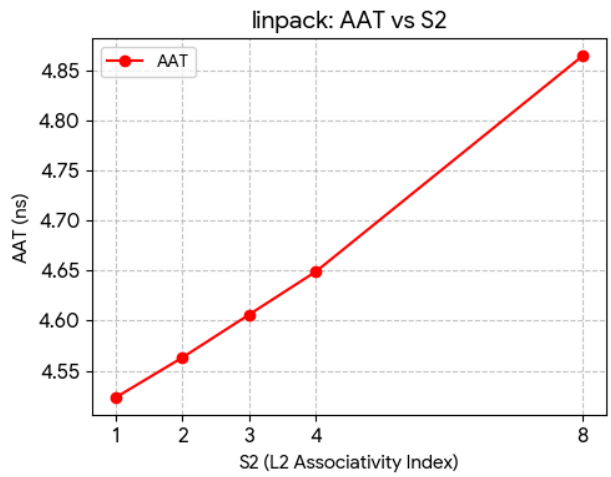
For the **linpack** trace, the best configuration is **L1(15, 7, 0) L2(17, 7, 1) MIP insertion policy** and **Plus1 prefetcher**, achieving an AAT of **4.523**.

Selection Rationale

Given that the metadata storage is identical across all 14 cases, following the principle of diminishing returns, the **Plus1 prefetcher** was selected over the Hybrid model because it achieved the peak performance of **4.523** with significantly lower hardware complexity.

Diminishing Returns





The point of diminishing returns for S1 is reached at **S1 = 0**. Similar to the other benchmarks, the 0.2 ns hit-time penalty incurred by moving from a direct-mapped to a 2-way associative cache (S1=1) outweighs the minor reduction in L1 conflict misses. The point of diminishing returns for S2 is reached at **S2 = 1 (2-way)**

Matmul Naive Trace

For the **matmul_naive** benchmark, I identified **6** configurations yielding the minimum AAT of **5.084**

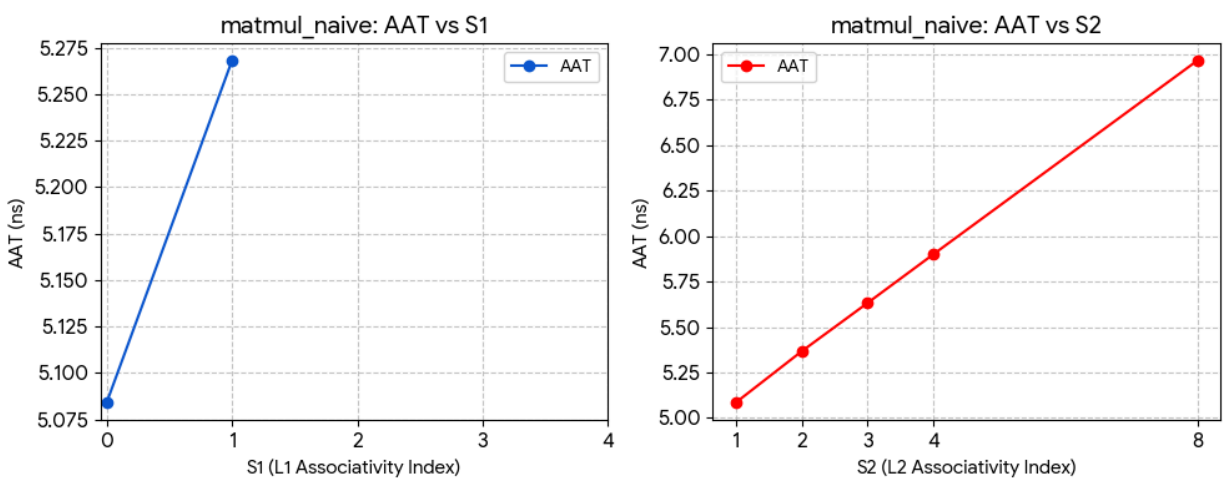
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Trace	C1	B	S1	C2	B	S2	L2 Policy	Prefetch	Rows	AAT	Data_B	Meta_B
2	matmul_naive	15	7	0	17	7	1	mip	hybrid	64	5.084	163840	64256
3	matmul_naive	15	7	0	17	7	1	mip	hybrid	256	5.084	163840	64256
4	matmul_naive	15	7	0	17	7	1	mip	hybrid	512	5.084	163840	64256
5	matmul_naive	15	7	0	17	7	1	lip	hybrid	64	5.084	163840	64256
6	matmul_naive	15	7	0	17	7	1	lip	hybrid	256	5.084	163840	64256
7	matmul_naive	15	7	0	17	7	1	lip	hybrid	512	5.084	163840	64256

For the matmul_naive benchmark, the best configuration is **L1(15, 7, 0) L2(17, 7, 1)**, **MIP insertion policy** and **Hybrid prefetcher** with **64 rows**, achieving an AAT of **5.084**

Selection Rationale

Following the principle of diminishing returns, the **Hybrid prefetcher** with **64 rows** was selected as it is the smallest predictor table size that achieves the peak performance. Increasing the prefetcher size to 256 or 512 rows provided no additional reduction in AAT.

Diminishing Returns



The point of diminishing returns for S1 is reached at **S1 = 0**. The point of diminishing returns for S2 is reached at **S2 = 1 (2-way)**.

Matmul Tiled Trace

For the **matmul_tiled** benchmark, I identified **6** configurations yielding the minimum AAT of **2.308**

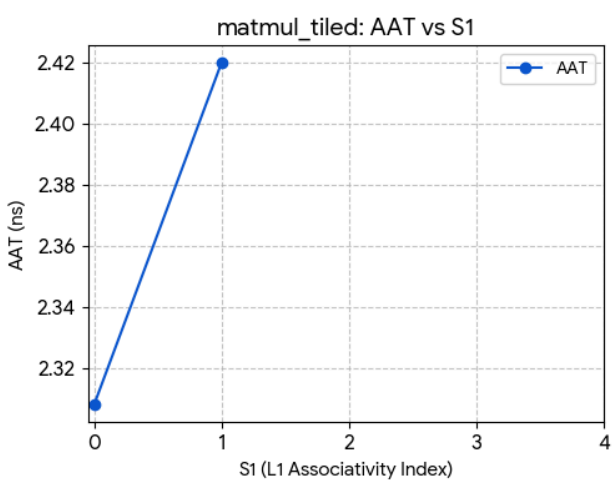
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	**Trace**	**C1**	**B**	**S1**	**C2**	**B**	**S2**	**L2 Policy**	**Prefetch**	**Rows**	**AAT**	**Data_B**	**Meta_B**
2	matmul_tiled	15	6	0	17	6	1	mip	hybrid	128	2.308	163840	128512
3	matmul_tiled	15	6	0	17	6	1	mip	hybrid	256	2.308	163840	128512
4	matmul_tiled	15	6	0	17	6	1	mip	hybrid	512	2.308	163840	128512
5	matmul_tiled	15	6	0	17	6	1	lip	hybrid	128	2.308	163840	128512
6	matmul_tiled	15	6	0	17	6	1	lip	hybrid	256	2.308	163840	128512
7	matmul_tiled	15	6	0	17	6	1	lip	hybrid	512	2.308	163840	128512

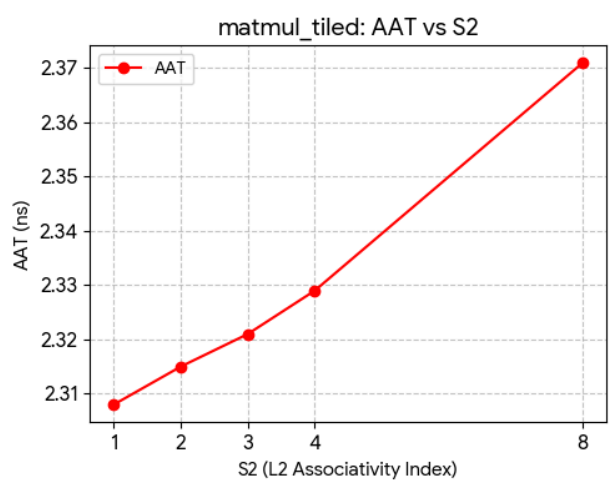
For the **matmul_tiled** benchmark, the best configuration is **L1(15, 6, 0), L2(17, 6, 1)**, **MIP policy** and **Hybrid prefetcher** with **128 rows**, achieving an AAT of **2.308**

Selection Rationale

Following the principle of diminishing returns, the **Hybrid prefetcher** with **128 rows** was selected as the optimal predictor size, as increasing the table to 256 or 512 rows yielded no further performance gain.

Diminishing Returns





The point of diminishing returns for S1 is reached at **S1 = 0**. The point of diminishing returns for S2 is reached at **S2 = 1 (2-way)**.

MCF Trace

For the mcf benchmark, I identified **14** configurations yielding the absolute minimum AAT of 2.209

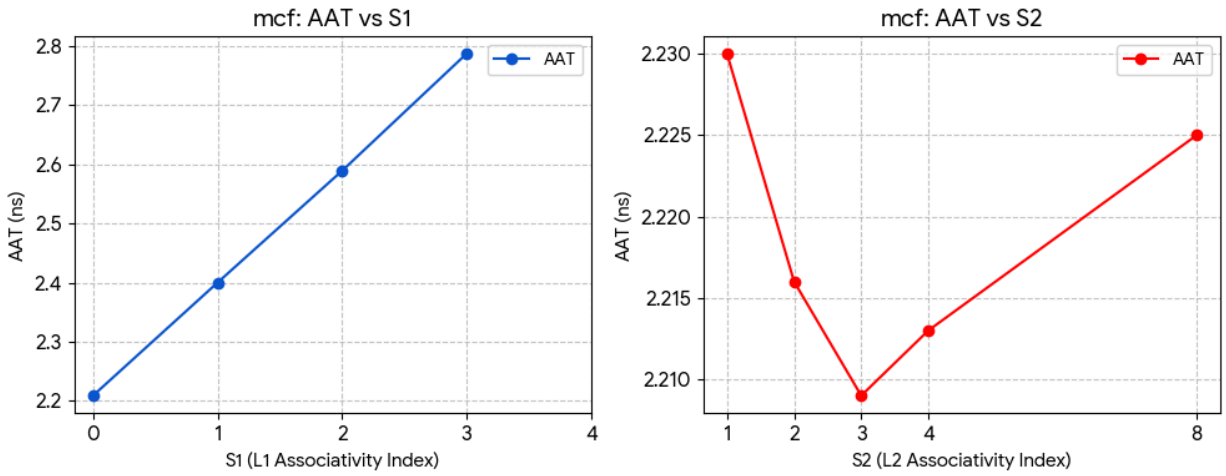
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	**Trace**	**C1**	**B**	**S1**	**C2**	**B**	**S2**	**L2 Policy**	**Prefetch**	**Rows**	**AAT**	**Data_B**	**Meta_B**
2	mcf	15	6	0	17	6	3	mip	plus1	0	2.209	163840	132608
3	mcf	15	6	0	17	6	3	mip	hybrid	16	2.209	163840	132608
4	mcf	15	6	0	17	6	3	mip	hybrid	32	2.209	163840	132608
5	mcf	15	6	0	17	6	3	mip	hybrid	64	2.209	163840	132608
6	mcf	15	6	0	17	6	3	mip	hybrid	128	2.209	163840	132608
7	mcf	15	6	0	17	6	3	mip	hybrid	256	2.209	163840	132608
8	mcf	15	6	0	17	6	3	mip	hybrid	512	2.209	163840	132608
9	mcf	15	6	0	17	6	3	lip	plus1	0	2.209	163840	132608
10	mcf	15	6	0	17	6	3	lip	hybrid	16	2.209	163840	132608
11	mcf	15	6	0	17	6	3	lip	hybrid	32	2.209	163840	132608
12	mcf	15	6	0	17	6	3	lip	hybrid	64	2.209	163840	132608
13	mcf	15	6	0	17	6	3	lip	hybrid	128	2.209	163840	132608
14	mcf	15	6	0	17	6	3	lip	hybrid	256	2.209	163840	132608
15	mcf	15	6	0	17	6	3	lip	hybrid	512	2.209	163840	132608

For the **mcf** benchmark, the best configuration is **L1(15, 6, 0) L2(17, 6, 3)**, **MIP policy** and **Plus1 prefetcher**, achieving an AAT of **2.209**

Selection Rationale

Following the principle of diminishing returns, the Plus1 prefetcher was selected over the Hybrid model because it achieved identical performance with significantly lower architectural complexity. Within the tied results, MIP was chosen over LIP as the industry-standard insertion policy.

Diminishing Returns



The point of diminishing returns for S1 is immediately at **S1 = 0.** For the L2 cache, the curve follows a more traditional "U-shape". The point of diminishing returns for S2 is **S2 = 3 (8-way)**